Jin Zhu

Ph.D. Student in Quantitative Biosciences | Georgia Institute of Technology

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Education

Georgia Institute of Technology Aug 2025–present
Ph.D. student, Quantitative Biosciences (Physics), School of Physics
Advisor: Dr. Peter J. Yunker

Shanghai Jiao Tong University Sep 2023–Jul 2025
M.S., Mechanical Engineering, UM–SJTU Joint Institute
Advisor: Dr. Zijie Qu | Thesis: *Bacterial Locomotion in Swimming and Swarming Motilities*

Shanghai Jiao Tong University Sep 2019–Aug 2023
B.Eng., Mechanical Engineering, UM–SJTU Joint Institute

Research Experience

Yunker Lab, Georgia Institute of Technology Aug 2025–present
Advisor: Dr. Peter J. Yunker

Hammer Lab, Georgia Institute of Technology Jan–May 2026
Rotation | Advisor: Dr. Brian K. Hammer

Bio-inspired Fluid Mechanics Lab, Shanghai Jiao Tong University Jan 2022–Jul 2025
Research Assistant | Advisor: Dr. Zijie Qu

LUIS Lab, Shanghai Jiao Tong University Jan–Sep 2022
Research Assistant | Advisor: Dr. Xuyang Lu

Publications

Name in bold; *equal contribution.

Journal articles

1. Bian, H., Li, C., Lin, Z., **Zhu, J.**, Chen, W., Li, G., Huang, Y., & Qu, Z. (2026). Active turbulence in shear-thinning fluid. *Physical Review Research* **8**, 013318. doi: 10.1103/z22f-3v22.
2. Li, C., Bian, H., Qiao, Y., **Zhu, J.**, & Qu, Z. (2025). Hydrodynamic interaction leads to the accumulation of *Chlamydomonas reinhardtii* near a solid–liquid interface. *Physical Review E* **111**, L052401. doi: 10.1103/PhysRevE.111.L052401. Editor's Suggestion.
3. **Zhu, J.***, Qiao, Y.*, Yan, L., Zeng, Y., Wu, Y., Bian, H., Huang, Y., Ye, Y., Huang, Y., Hii, R. C. W., Teng, Y., Guo, Y., Li, G., & Qu, Z. (2025). Propulsion contribution from individual filament in a flagellar bundle. *Applied Physics Letters* **126**, 073702. doi: 10.1063/5.0243416.

Conference proceedings

4. Cai, R., Niazi, M. D. K., Ai, Y., **Zhu, J.**, Wu, L., & Lu, X. (2025). Fully integrated self-propelling microrobot in 180 nm CMOS with sub-GHz parity-time-symmetry on-chip energy harvesting and traveling wave electroosmosis actuation. *IEEE International Solid-State Circuits Conference (ISSCC)*, pp. 1–3. doi: 10.1109/ISSCC49661.2025.10904559.

Preprints

5. Lin, Z.*, **Zhu, J.***, Bian, H., & Qu, Z. (2026). Hydrated porous substrates modulate the spatiotemporal dynamics of bacterial

turbulence. *arXiv*. arXiv:2611.04668. Under review.

6. Gu, P., Lin, Z., Ren, H., **Zhu, J.**, Li, A., Said, O., Chen, W., & Qu, Z. (2026). Inclined surfaces bias bacterial swarming expansion and edge organization. *arXiv*. arXiv:2610.01281. Under review.
7. Krueger, A., Bogati, B., Copeland, R., Hejri Bidgoli, M. S., **Zhu, J.**, Weiss, D. S., & Yunker, P. J. (2026). Rapid bacterial identification through biophysical surface topography features. *bioRxiv*. doi: 10.64898/2026.03.12.711416. Under review.
8. Duan, S., Wang, Z., Xiong, K., **Zhu, J.**, Gu, P., Chen, W., Xin, H., & Qu, Z. (2026). Population-scale advancing interface modeling reveals how bacterial swarms encode future spatial architecture. *arXiv*. arXiv:2602.01056. Under review.

Selected Presentations

78th Annual Meeting of the APS Division of Fluid Dynamics 2025
Houston, TX. Oral presentation: *Complex Porous Surface Modulate Bacterial Turbulence.*
Poster: *Propulsion Contribution from Individual Filament in a Flagellar Bundle.*

Chinese Congress of Theoretical and Applied Mechanics 2025
Changsha, China. *Complex Porous Surface Modulate Bacterial Turbulence.*

13th Chinese Fluid Mechanics Academic Conference 2024
Harbin, China. *Study on Microbial Flagellar Bundling Process and Propulsive Force Generation Based on Macroscopic Experimental Model.*

Teaching Experience

Teaching Assistant, School of Physics, Georgia Tech Aug 2025–May 2026
PHYS 2212: Principles of Physics II (Fall 2025; Spring 2026).

Teaching Assistant, UM–SJTU Joint Institute 2021–2024
VM320: Fluid Mechanics; VM312: Mechanical Behaviour of Biological Tissues;
VM250: Design and Manufacturing I; VG100: Introduction to Engineering.

Honors, Fellowships & Leadership

InQuBATE Fellowship, Georgia Institute of Technology Aug 2026–present
QBioS SGA, Georgia Institute of Technology July 2026–present
JS Weitz Excellency in Travel Funds, Georgia Institute of Technology 2025
Outstanding Graduate of Shanghai 2023
ChunTsung Scholar, Shanghai Jiao Tong University 2023
Capstone Project Gold Award, UM–SJTU Joint Institute Design Expo 2022
John Wu & Jane Sun Merit Scholarship 2022
Shanghai Jiao Tong University B-level Scholarship 2022
First Prize, Shanghai Mechanical Engineering Innovation Competition 2021
Co-founder and Co-leader, ASME Branch, UM–SJTU Joint Institute 2022–2024

Technical Skills

Programming: Python, MATLAB, C/C++
Modeling & analysis: COMSOL, SolidWorks, LabVIEW, Origin
Fabrication: Arduino, 3D printing (PLA, TPU, resin), laser cutting
Laboratory: Bacterial culture, PCR
Writing & graphics: LaTeX, Adobe Illustrator